GitHub and GitHub Desktop

International Marketing Analytics

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Today: GitHub from zero to your own repository

Account → Education → Install → Clone → Pull and push → Private or public

Set up

- A GitHub account with two-factor sign-in
- Your e-mail kept private
- GitHub Education for free student benefits

Work

- GitHub Desktop installed and signed in
- Clone the course repository
- Fetch, pull, commit, push

Share

- Your own repository, private by default
- Invite your group members
- When public makes sense, and when it does not

No command line: every step in this guide is a click in GitHub Desktop or on github.com.

Git, GitHub and GitHub Desktop

Git

The engine: records every saved version of a folder

GitHub

The website: keeps a copy online and lets others access it

GitHub Desktop

The app: runs Git with buttons and brings its own Git

Word	Meaning	Button in GitHub Desktop
Repository (repo)	a project folder with its full history	Current repository
Clone	download a repo to your laptop, linked to GitHub	File → Clone repository
Commit	save a snapshot with a short message	Commit to main
Push / pull	send your commits up / get new commits down	Push origin / Pull origin
Fetch	check GitHub for new commits, change nothing	Fetch origin

Your GitHub account

Part 1 of 6 · sign up, two-factor sign-in, a private e-mail address

Create your GitHub account

1. Open github.com/signup
2. **E-mail:** your WU student address
3. **Password** of at least 15 characters, or 8 with a number and a lower-case letter
4. **Username:** short and professional, e.g. anna-berger; it is public
5. **Verify:** type the code GitHub sends to your inbox
6. **Plan:** choose Free; skip the questions if you like

Good to know

- One account per person: keep it for life
- Free plan: unlimited public and private repos
- Your username appears on every commit

Have an account? Keep it; add your WU address under Settings → Emails.

Secure it: two-factor authentication (2FA)

Turn on 2FA

1. **Profile picture** → Settings
2. **Password and authentication** → Enable two-factor authentication
3. **Scan the QR code** with an authenticator app
4. **Type** the 6-digit code from the app
5. **Download** your recovery codes and keep them safe
6. **Add a backup:** GitHub Mobile or a passkey

45
days to set it up once GitHub asks

TOTP
app recommended as main method

16
recovery codes: your way back in

Apps that work: Microsoft Authenticator, Google Authenticator, 1Password, or GitHub Mobile as backup.

Keep your e-mail address private

In your GitHub settings

1 **Settings** → Emails

2 **Add** your WU address if it is missing, and verify it

3 **Tick** Keep my email addresses private

4 **Tick** Block command line pushes that expose my email

5 **Copy** the noreply address shown there

Your noreply address (example)

```
1234567+anna-berger@
  users.noreply.github.com
```

Public commits show the commit e-mail to anyone

The noreply address still links commits to you

WU address: keep it verified on your account. GitHub Education checks it, even when your commits use the noreply address.

GitHub Education

Part 2 of 6 · who qualifies, what you get, how to apply

GitHub Education: what students get

GitHub Copilot

- Free for verified students, with premium features
- Works in Positron and GitHub Desktop

Codespaces

- Cloud computers: up to 180 core hours a month
- Not needed for this course

Developer Pack

- Free offers from partner companies
- Tools, domains, learning platforms

Portal: education.github.com shows your benefits once you are verified

Time-limited: GitHub re-checks your status; benefits end when you are no longer a student

Sources: GitHub Docs, About GitHub Education for students (2026)

Who qualifies, and which proof works

You qualify if you

- Study in a degree programme (WU: yes)
- Have your own personal GitHub account
- Are at least 13 years old
- Can prove current enrolment

Proof GitHub accepts

- Student ID showing the current enrolment date
- Enrolment confirmation (Studienbestätigung)
- Transcript of records
- Your class schedule

Academic e-mail first: add and verify your WU address in GitHub before you apply; lower case, no plus sign (anna+x@...).

Sources: GitHub Docs, Apply to GitHub Education as a student (2026)

Apply in six steps

1. Open github.com/settings/education/benefits
2. **Click** Start an application; choose Student
3. **School:** Vienna University of Economics and Business
4. **E-mail:** select your verified WU address
5. **Proof:** upload or photograph your document
6. **Submit** and wait for the e-mail with the decision

Minutes to several days: carry on with Part 3 meanwhile.

Photo tips

- Whole document in the frame
- Name, school and current date readable
- Daylight, no glare
- Allow camera and location if the browser asks
- Check the status on the same page

Not approved? Common reasons and fixes

Reason	What to do
Blurry document, or no current date	Reapply with a sharp photo of a dated enrolment confirmation
WU address not added or not verified	Settings → Emails: add it, click the link in the confirmation e-mail
Address with capitals or a plus sign	Delete it, add it again in lower case without +
Address already used by another account	Use the account that got the pack; merge accounts
Still stuck	support.github.com/contact/education, describe each step you tried

Reapply any time: same page, Start an application.

Install and set up GitHub Desktop



Part 3 of 6 · download, sign in, your name and e-mail for commits

Install GitHub Desktop

Windows 10/11 (64-bit)

1. Open desktop.github.com
2. **Click** Download for Windows
3. **Double-click** the setup file in Downloads
4. **Wait:** GitHub Desktop opens by itself

macOS 12 or later

1. Open desktop.github.com
2. **Click** Download for macOS
3. **Double-click** the zip file, then drag the app to Applications
4. **Open** GitHub Desktop from Applications

Git included: you do not need to install Git separately

Updates: GitHub Desktop updates itself; restart it when it says so

Sign in to GitHub in GitHub Desktop

Open the Accounts pane

1. **Windows:** File → Options → Accounts
2. **macOS:** GitHub Desktop → Settings → Accounts
3. **Click** Sign in to GitHub.com
4. **Click** Continue with browser

In the browser

1. **Sign in** with username and password
2. **Enter** the 2FA code from your app
3. **Click** Authorize desktop
4. **Allow** the browser to open GitHub Desktop

Done when: your name and avatar appear in the Accounts pane.

Your name and e-mail for commits

Options / Settings → Git

1. **Windows:** File → Options → Git
2. **macOS:** GitHub Desktop → Settings → Git
3. **Name:** your real name, e.g. Anna Berger
4. **Email:** pick the …@users.noreply.github.com address
5. **Default branch:** leave it at main; click Save

Name

```
Anna Berger
```

Email

```
1234567+anna-berger@users…
```

Default branch

```
main
```

Privacy: with the noreply address your WU or private e-mail never shows in a public repository.

Clone a repository

Part 4 of 6 · two ways to clone, where to put the folder, opening it in Positron

Clone: two ways, same result

A From github.com

1. **Open** the repository page on github.com
2. **Click** the green Code button
3. **Click** Open with GitHub Desktop
4. **Choose…** a local folder, then Clone

B In GitHub Desktop

1. **File →** Clone repository
2. **Tab URL:** paste the repository address
3. **Local path:** Choose… a folder
4. **Click** Clone

The course repository (public: no sign-in needed to clone)

```
https://github.com/arnefloh-wu/international-marketing-analytics-teaching
```

Where the folder should live

Good places

- Windows: C:\Users\anna\wu\ima
- macOS: /Users/anna/wu/ima
- Short path, no spaces, no umlauts
- On the laptop's own disk

Avoid

- OneDrive, iCloud Drive, Dropbox folders
- Desktop and Documents if they sync to the cloud
- Paths like …/Mein Kurs (WU)/
- USB sticks and network drives

Find the folder later: Repository → Show in Explorer (Windows) / Show in Finder (macOS)

Moved it by mistake? GitHub Desktop says it cannot find it: click Locate… and pick the new place

Open the clone in Positron

Clone in GitHub Desktop → **Positron: Open Folder** → **Pick the cloned folder** → **Work, save** → **Back to GitHub Desktop**

Positron

Edit and run code, render Quarto files.

Saved changes show up in GitHub Desktop at once.

GitHub Desktop

Fetch, pull, commit and push.

Keep it open next to Positron.

Tip: under Options → Integrations you can set an external editor; if Positron is not listed, simply open the folder from Positron.

Pull, commit, push

Part 5 of 6 · the everyday cycle, undoing mistakes, what never goes in

The everyday cycle

1 Fetch origin → 2 Pull origin → 3 Work in Positron → 4 Commit to main → 5 Push origin

Fetch origin	asks GitHub what is new; changes nothing on your laptop
Pull origin ↓ 3	downloads 3 new commits into your folder
Commit to main	saves a snapshot of your checked files, on your laptop only
Push origin ↑ 1	uploads your 1 new commit to GitHub

Course repository: only fetch and pull, before every session. Your own work goes into your own folder or repository.

Pull: get the latest version

Before every session

1. **Current repository:** pick the course repository
2. **Click** Fetch origin
3. **If it shows Pull origin ↓:** click it
4. **Done:** the button says Fetch origin again

Pull refused?

- You edited a file that the update changes
- Changes tab: right-click it → Discard changes
- Then click Pull origin again

Golden rule: never edit files inside the course repository. Copy them into your own project folder first.

Commit: save a snapshot

In the Changes tab

1 **Look** at the list of changed files

2 **Click a file** to see the diff: red removed, green added

3 **Tick** the files that belong together

4 **Summary:** what changed, e.g. Add Austria sales chart

5 **Click** Commit to main

Good commits

- Small: one step per commit
- Summary in the imperative: Add, Fix, Update
- Commit when something works
- Copilot button: draft message, then check

History tab: every commit with author, date and diff. You can always go back.

Push: upload your commits

Push

1 **After committing, click** Push origin ↑

2 **New commits on remote?** click Fetch, then Pull origin

3 **Then** Push origin again

4 **Check** on github.com: Repository → View on GitHub

100 MiB
largest file GitHub accepts

50 MiB
GitHub warns from this size

Asked to create a fork? you tried to push to a repository you cannot write to, such as the course repository. Cancel, and keep your work in your own repository.

Oops: undo and fix

Situation	In GitHub Desktop
Changed a file, want the old version	Changes tab: right-click the file → Discard changes
Last commit was wrong, not pushed yet	Click Undo next to the commit message
A pushed commit was wrong	History: right-click the commit → Revert changes in commit
Merge conflict after a pull	Open the file, keep the right lines, delete <<<<<<< ======= >>>>>>>, commit
Folder moved or renamed	Click Locate… and pick the folder

Lost? Do not delete the folder. Ask first: almost everything can be recovered.

What never goes into a repository

Secrets

Passwords, API keys, tokens

Files such as .env

Personal data

Names, e-mails, survey answers

Licensed or confidential data

Big or generated

The .venv folder

Files over 50 MiB, rendered _files

.gitignore (example)

```
.venv/
.env
__pycache__/
*_files/
.DS_Store
```

Ignore a file: Changes tab → right-click → Ignore file

Edit the list: Repository → Repository settings → Ignored files

Pushed a secret? Change the password or key at once

Private or public?

Part 6 of 6 · your own repository, inviting your group, changing visibility

Private and public repositories

	Private	Public
Who can see it	you and people you invite	everyone on the internet
Who can clone it	you and invited people	anyone, without an account
Who can push	you and invited collaborators	you and invited collaborators
Cost on GitHub Free	free, unlimited	free, unlimited
Use it for	group work, assignments, drafts	portfolio, open-source, the course materials

Rule for this course: keep group and assignment repositories private. Never publish data you received under licence or solutions others could copy.

Create your own repository and publish it

1 Create on your laptop

1. **File →** New repository
2. **Name:** e.g. ima-group-3, no spaces
3. **Local path:** Choose… your wu folder
4. **Tick** Initialize with a README
5. **Git ignore:** Python; click Create

2 Publish to GitHub

1. **Click** Publish repository
2. **Check** name and description
3. **Keep this code private:** leave it ticked
4. **Organization:** None (your account)
5. **Click** Publish repository

Existing folder? File → Add local repository, create a repository there, then publish.

Invite your group members

Owner: on github.com

1. **GitHub Desktop:** Repository → View on GitHub
2. **Click** Settings (the repository's, top right)
3. **Left menu:** Collaborators
4. **Click** Add people, type the username
5. **Click** Add … to this repository

Members

- Accept the invitation: e-mail or github.com/notifications
- Clone the repository (Part 4)
- Pull before you start, push when you stop
- One person per file where possible

Invitations expire after seven days; the owner can resend them.

Change visibility later

On github.com

1. **Open** the repository → Settings
2. **General:** scroll to the Danger Zone
3. **Click** Change visibility
4. **Confirm** with the repository name

Before you go public

- The whole history becomes visible, not just today's files
- Anyone can copy and fork it, for good
- Check: no secrets, no personal or licensed data

Public to private: stars and watchers are erased, and copies others already forked stay public.

When something goes wrong

Message or problem	Fix
Authentication failed / Bad credentials	Options or Settings → Accounts: sign out, sign in again
Push rejected: new commits on remote	Fetch origin, Pull origin, then Push origin
File over 100 MiB	Remove it from the commit; add it to .gitignore
Clone failed in OneDrive or a redirected folder	Clone into a local folder such as C:\Users\you\wu
macOS: username or passphrase not correct	Keychain Access: lock, then unlock 'login'; sign in again
Cannot find the repository	Click Locate… and choose the folder

Log files: Help → Show Logs in Explorer (Windows) or Show Logs in Finder (macOS); send them when you ask for help.

Checklist

1. **Account** with 2FA and recovery codes stored
2. **E-mail:** WU address verified, Keep my email addresses private on
3. **GitHub Education** application submitted
4. **GitHub Desktop** installed, signed in, noreply address under Git
5. **Course repository** cloned to a local folder and opened in Positron
6. **Group repository** created, private, members invited (optional)

Official guides

- GitHub Desktop: getting started
- Cloning a repository in GitHub Desktop
- Committing, pushing and syncing
- Two-factor authentication
- Apply to GitHub Education as a student
- Setting repository visibility
- GitHub Skills: free interactive courses

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